Introduction to the Laws of Motion

Lesson 1: What is Motion?

What is Motion? – Study Notes

Key Concepts

1. **Motion** – Change of an object’s position with respect to a reference point over time.
2. **Speed** – How fast an object moves; a scalar quantity (only magnitude).
3. **Velocity** – Speed with a specified direction; a vector quantity.
4. **Reference Frame** – The point of view (e.g., a stationary observer on the ground) used to measure motion.
5. **Uniform vs. Non uniform Motion** – Motion at constant speed/direction versus motion that changes speed or direction.

Summary

Motion is a fundamental concept in physics that describes how an object’s location changes over time. To talk about motion we must first choose a **reference frame**—a fixed point or object from which we measure positions. Once a reference frame is set, we can quantify motion using **speed** and **velocity**. Speed tells us *how fast* something is moving, ignoring direction, while velocity tells us both *how fast* and *in which direction* the object travels.

In everyday life, we encounter both uniform (constant speed, straight line) and non uniform motion (speed or direction changes). Understanding the distinction between scalar (speed) and vector (velocity) quantities is essential because it determines how we add, subtract, or compare different motions. Simple observations—like a ball rolling across a floor—illustrate these ideas and lay the groundwork for more advanced topics such as acceleration, forces, and kinematics.

Important Points

- **Motion** = change in position **9G,¢¢ ðver a time interval **9GB¢¢à
- **Speed** = \(\displaystyle \text{speed}= \frac{\text{distance traveled}}{\text{time}}\) (units: m /s {^{1}, km /h {^{1}, etc.).
- **Velocity** = \(\displaystyle \vec{v}= \frac{\vec{\Delta x}}{\Delta t}\) (units same as speed, but includes direction).
- Speed is a **scalar**; it has magnitude only.
- Velocity is a **vector**; it has magnitude *and* direction.
- The same speed can correspond to many different velocities depending on direction.
- **Reference frame** matters: an object may be at rest in one frame and moving in another.
- **Uniform motion** !’ constant speed and direction !’ straight line path.
- **Non uniform motion** !’ either speed or direction (or both) changes !’ curved or accelerating path.

Examples

Example 1 – Rolling Ball (Uniform Motion)

A ball rolls 10 /m straight across a hallway in 5 /s.

- **Speed:** \(\frac{10\ \text{m}}{5\ \text{s}} = 2\ \text{m /s}^{-1}\).
- **Velocity:** If the ball moves east, \(\vec{v}=2\ \text{m /s}^{-1}\,\hat{i}\) (east direction).

The ball’s speed and velocity are constant !’ uniform motion.

Example 2 – Ball Bouncing Down a Stairs (Non uniform Motion)

A ball rolls down a set of stairs, covering 12 /m in 4 /s, but it speeds up as it descends.

- Average speed = \(\frac{12\ \text{m}}{4\ \text{s}} = 3\ \text{m /s}^{-1}\).
- Because the ball’s speed is increasing, its **instantaneous velocity** changes during the motion, making it non uniform.

Quick Review

1. **What is the difference between speed and velocity?**
2. **Why do we need a reference frame when describing motion?**
3. **If a car travels 60 /km north in 1 /h, what is its speed and its velocity?**
4. **Define uniform motion in one sentence.**
5. **Give a real world example of non uniform motion and explain why it fits the definition.**

Further Reading

- **Acceleration** – how velocity changes with time.
- **Newton’s First Law** – inertia and the concept of an object at rest or in uniform motion.
- **Vector addition** – combining multiple velocities.
- **Graphs of motion** – position time and velocity time plots.
- **Reference frames in relativity** – what changes when speeds approach the speed of light.